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RESEARCH ARTICLE

A Journey From Traditional Supply Chain Processes to Sustainability-Oriented Blockchain Supply Chain: The Critical Role of Organizational Capabilities

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Received: 24 April 2024 | **Revised:** 20 November 2024 | **Accepted:** 10 January 2025

Keywords: blockchain technology | electronics companies | environmental sustainability | firm performance | SEM | sustainable development

ABSTRACT

Blockchain technology has a great potential to transform how businesses approach sustainability by offering transparency, efficiency, and accountability across various processes. By adopting blockchain, businesses can not only meet growing consumer and regulatory demands for sustainability but also create competitive advantages by promoting resource efficiency, fostering trust, and building eco-conscious operations. Building on dynamic capabilities lens within the resource-based view (RBV) framework, and in order to know about antecedents and consequences of blockchain technology adoption, this study investigates how three types of organizational capabilities—namely, innovation capability, information-sharing capability, and technological capability—may facilitate the blockchain technology adoption and most importantly, how such adoption could impact firm performance. To test our research hypotheses, we collected survey data from 198 electronics companies in Dhaka, Bangladesh. The structural equation modelling has been used to test the model hypothesis. The findings showed that all three organizational capabilities—innovation capability, information-sharing capability, and technological capability—significantly facilitate blockchain technology adoption, which in turn leads to enhanced firm performance in small- and medium-sized enterprises, demonstrating the strategic value of blockchain in driving sustainability and competitiveness in electronics companies.

1 | Introduction

In the face of rapid technological change and growing sustainability demands, electronics companies and other industrial sectors must develop adaptable capabilities to enhance resilience and competitiveness while addressing environmental challenges (Lee and Pilkington 2017). Although the resource-based view (RBV) has traditionally emphasized static resources—those assets and competencies that remain stable over time (Barney 1991; Lockett, Thompson, and Morgenstern 2009; M. Kim, Song, and Triche 2015)—the accelerating pace of innovation calls for a dynamic capabilities perspective. This perspective highlights the need for firms to continuously reconfigure and deploy resources to navigate emerging challenges and evolving market demands (Ambrosini, Bowman, and Collier 2009; Barreto 2010; Teece 2012). The adoption of technologies like blockchain, which offers both operational efficiency and environmental benefits, has become essential for companies aiming to remain relevant while addressing increasing sustainability concerns (Nowotny et al. 2018; S. Zhang et al. 2023; Eslami et al. 2024). To address this theoretical gap, our study applies a dynamic capabilities lens within the RBV framework to examine how electronics companies can reconfigure their resources to

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integrate blockchain into their supply chains as a sustainable innovation. Therefore we attempt to know *what organizational capabilities are most critical for blockchain adoption as an environmentally friendly practice, and how might this adoption shape the performance of electronics companies?*

In recent years, the growing challenges posed by climate change and environmental hazards have made sustainability-oriented practices not merely a strategic choice but a necessity for electronics companies that are striving to mitigate risks and ensure long-term viability (Zaid, Jaaron, and Bon 2018; Yildiz Çankaya and Sezen 2019; Torelli, Balluchi, and Lazzini 2020; Jraisat et al. 2023; Ghag and Sonar 2024). These environmentally friendly practices are crucial as they directly address the environmental impact of corporate activities and encompass areas such as production, resource utilization, and waste management (Teece 2012; Jraisat et al. 2023). Sustainability initiatives play an essential role in combating climate change by emphasizing responsible resource use, waste reduction, and energy efficiency and minimizing carbon emissions. Increasing attention has been directed towards understanding the factors that drive businesses to adopt sustainability-oriented practices and how such adoption influences subsequent firm performance (Nayal et al. 2022; Srivastava et al. 2022; Samadhiya et al. 2023; Costa Melo et al. 2023; Joshi and Sharma 2023; Belhadi et al. 2024; Kazançoğlu et al. 2024). This focus has intensified as global concerns such as resource depletion and climate change push electronics companies to rethink and restructure their operational strategies (S. Kim, Lee, and Lim 2021; Sharma et al. 2022; Sahoo, Kumar, and Upadhyay 2023; Muhamed et al. 2023; Shishodia, Sharma, Pavithra, and Mathiyazhagan 2023; Torelli and Balluchi 2023; Chowdhury et al. 2024).

In response, blockchain technology has emerged as a transformative tool for electronics companies by offering a decentralized, immutable ledger that enhances the tracking of products, resources, and energy consumption throughout supply chains (Sharma et al. 2022; Chowdhury et al. 2024). This heightened transparency enables electronics companies to monitor their environmental impact at each stage from sourcing raw materials to product distribution (Samadhiya et al. 2023; Kazançoğlu et al. 2024). By adopting blockchain, businesses can develop comprehensive systems that trace products from creation to disposal or recycling, thereby reducing environmental harm and promoting more sustainable practices (Ghag and Sonar 2024). Blockchain technology makes businesses greener by helping them to identify inefficient and environmentally harmful practices within supply chains (Kazançoğlu et al. 2024). Companies adopting blockchain can ensure more sustainable logistics, reducing unnecessary transportation and energy consumption, thereby contributing to business sustainability (Samadhiya et al. 2023; Ghag and Sonar 2024).

While much of the existing literature on blockchain adoption has focused on developed markets such as South Korea (Lee and Pilkington 2017), Germany (Xu et al. 2024), the United States (Queiroz and Wamba 2019), Canada (Ducas and Wilner 2017), and Japan (Ahl et al. 2020)—where infrastructure, resources, and regulatory frameworks are more robust—there is limited research addressing the unique challenges and opportunities faced by companies in developing economies like Bangladesh (Bhattacharjee, Jahanshahi, and Chakraborty 2024). Firms in these markets often operate in resource-constrained environments with evolving regulatory landscapes, requiring them to mobilize and adapt their internal capabilities to adopt new technologies successfully (Seman et al. 2019; Dimple and Tripathi 2024). Beyond resource limitations, the successful adoption of blockchain technology largely depends on organizational capabilities that enable effective implementation and integration of such complex systems (Teece 2012; Ghag and Sonar 2024). Addressing this contextual research gap, our study applies the lens of dynamic capabilities within the RBV framework. We predict that three key organizational capabilities—innovation, information sharing, and technological capability—are essential for firms to continuously adapt to changes and integrate blockchain technology effectively into their operations.

Firstly, blockchain technologies are quite new and require electronics companies to innovate and invest more on research and development (R&D) projects and initiatives (Kazançoğlu et al. 2024). Business with more financial capabilities to invest on innovative project are better equipped to experiment with and adopt blockchain solutions, as they have the organizational mindset and resources to explore new technologies and creatively apply them to their operations (Teece 2012; Belhadi et al. 2024). Secondly, blockchain's value mainly lies in its ability to decentralize and secure data (Samadhiya et al. 2023), but for it to function effectively, businesses must have robust and strong systems for sharing and managing information internally (Yildiz Çankaya and Sezen 2019). Electronics companies that already excel at effectively sharing information across internal departments or external partners are better positioned to benefit from blockchain's transparency and security features. Lastly, blockchain is a sophisticated technology that demands a significant technological infrastructure and skilled personnel for its adoption and maintenance over time (Costa Melo et al. 2023). Electronics companies with strong technological capabilities, such as advanced IT infrastructure, technical expertise, and on-going investments in technology are in better strategic position to handle technical challenges related to blockchain technology integration (Shishodia, Sharma, Pavithra, and Mathiyazhagan 2023). The business technological capability enables them to more effectively overcome the complexities of blockchain setup, customization, and scaling (Teece 2012).

The adoption of blockchain technology in supply chain management may improve electronics companies in different ways (Nayal et al. 2022; Srivastava et al. 2022; Samadhiya et al. 2023; Costa Melo et al. 2023; Joshi and Sharma 2023; Belhadi et al. 2024; Kazançoğlu et al. 2024). Conventional supply chain systems often face challenges such as fraud, counterfeiting, inefficiencies, and a lack of transparency. Blockchain, with its inherent features of immutability, security, transparency, and decentralization, offers a solution to these business issues (Feng et al. 2018; Seman et al. 2019). By enabling the real-time tracking of products throughout the supply chain, blockchain ensures end-to-end visibility. It records key information such as the origin, manufacturing processes, storage conditions, and transportation details, thus helping firms ensure product authenticity and quality (Habib, Bao, and Ilmudeen 2020). Moreover, blockchain facilitates proactive risk management by enabling real-time event detection through integrated technologies (Mardani

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et al. 2020). Lastly, the incorporation of blockchain into supply chains not only boosts operational efficiency but also aligns with the sustainability goals of firms, further enhancing performance (Cousins et al. 2019; B. Zhang et al. 2022; Dimple and Tripathi 2024).

While climate change and businesses' role in environmental degradation have gained significant attention in developed and transitional economy (Pimenova and Van Der Vorst 2004; Dean and McMullen 2007; Zheng, Luo, and Wang 2014), less focus has been placed on how firms in developing countries address these challenges (Nurunnabi 2016; Ajide, Soyemi, and Oladipupo 2024). Our research bridges this contextual gap by examining electronics companies in Bangladesh, a nation highly vulnerable to environmental hazards, ranked ninth globally in climate disaster risk, and burdened by severe air pollution. This context emphasizes the need for sustainable practices in emerging markets. Our study contributes to the literature by illustrating how Bangladeshi firms leverage organizational capabilities—innovation, information sharing, and technological capacity—to adopt blockchain as a sustainability-oriented technology. While most blockchain research has focused on developed markets (Teece 2012; Seman et al. 2019; Dimple and Tripathi 2024), our study sheds light on its adoption in an emerging economy where economic and regulatory conditions often differ. Specifically, innovation capability fosters a culture of adaptability and experimentation, critical for implementing disruptive technologies like blockchain. Information-sharing capability enhances data integrity and traceability across supply chains, enabling transparency and collaboration while aligning with environmental standards (Teece 2012; Dimple and Tripathi 2024). Finally, technological capability provides the skills and infrastructure needed to integrate and manage blockchain, overcoming common limitations in technical resources. By highlighting how firms in resource-limited settings mobilize these capabilities, we show how blockchain adoption can drive both sustainability and improved firm performance, addressing challenges unique to developing contexts (Barney 1991; Seman et al. 2019).

Furthermore, our study extends the RBV theory by integrating dynamic capabilities as a key mechanism through which firms reconfigure and adapt their internal resources (Barney 1991; Seman et al. 2019). While RBV has traditionally emphasized static resources as the primary source of competitive advantage, this study advances the theory by positioning dynamic capabilities as essential for responding to rapidly evolving technological and environmental pressures (Barney 1991; Seman et al. 2019). Blockchain adoption, as a contemporary and sustainable technology, requires firms to exhibit flexibility, adaptability, innovativeness, and a proactive approach to reconfiguring their capabilities (B. Zhang et al. 2022; Dimple and Tripathi 2024). By showing how organizations utilize dynamic capabilities—such as innovation, information sharing, and technological infrastructure—to integrate blockchain, we contribute to a more nuanced understanding of how firms can gain competitive advantage and enhance sustainability. This research not only reinforces the relevance of dynamic capabilities within the RBV framework but also provides new insights for both scholars and practitioners on how businesses can respond strategically to disruptive technologies in a resource-constrained environment (Barney 1991; Seman et al. 2019; B. Zhang et al. 2022; Dimple and Tripathi 2024).

2 | Literature Review and Hypothesis Development

Business operations in developing countries, such as Bangladesh, often contribute to significant environmental degradation due to inadequate regulatory frameworks and limited resources for sustainable practices (Seman et al. 2019; Dimple and Tripathi 2024). Conventional supply chains in these regions are characterized by complex logistics and multiple intermediaries, which lead to increased carbon emissions from transportation and energy consumption (Cousins et al. 2019; B. Zhang et al. 2022; Dimple and Tripathi 2024). Additionally, the lack of visibility in traditional supply chains makes it difficult to trace the origin and environmental impact of raw materials and products, often resulting in unethical sourcing practices, excessive carbon footprints, and inefficient resource use. Resource waste is further exacerbated by inefficiencies in production, transportation, and inventory management (B. Zhang et al. 2022; Dimple and Tripathi 2024). These harmful impacts underscore the urgent need for businesses to adopt sustainability-oriented practices. To mitigate these negative effects, many firms are turning to innovative technologies, like blockchain, which can enhance transparency, traceability, and efficiency across supply chains. This not only promotes environmentally responsible practices but also aligns business strategies with global sustainability goals.

As an example, companies like IBM used blockchain to track the origin of food, increasing transparency and reducing waste. Walmart leveraged it for food traceability, cutting waste and enhancing safety by swiftly identifying contamination sources. Loopring facilitates trading renewable energy certificates. Unilever uses blockchain for sustainable tea production, promoting fair trade and environmental protection. Similarly, De Beers tracks diamond origins through Tracr, ensuring integrity and reducing fraud. These examples show how blockchain helps businesses meet sustainability goals by increasing transparency, accountability, and efficiency in supply chains (Tumpa et al. 2019; Dimple and Tripathi 2024).

From a theoretical standpoint, the RBV (Feng et al. 2018; Seman et al. 2019) offers a robust framework for understanding how firms can effectively adopt and utilize new and modern technologies like blockchain. According to the RBV, electronics companies with valuable, rare, inimitable, and nonsubstitutable resources are better positioned to gain a competitive advantage (Barney 1991; Seman et al. 2019). In the context of blockchain adoption, some of these resources and capabilities include innovation capabilities, efficient information-sharing capabilities, and technological capabilities. Blockchain, as a complex and evolving technology, requires electronics companies to continually adapt and refine their organizational processes (Wamba, Queiroz, and Trinchera 2020). This study posits that a combination of these organizational capabilities will not only facilitate blockchain adoption but also improve firm performance, particularly in developing countries where resource constraints and environmental challenges

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are prevalent (Seman et al. 2019). In the next section, we first explain how each of these business capabilities directly improves firm performance. Second, we discuss how these capabilities indirectly contribute to business performance by facilitating the adoption of blockchain technology in supply chains (Urban and Matela 2022).

2.1 | Innovation Capability and Adoption of Blockchain Technology

Innovation capability refers to a company's ability to develop new ideas, products, or processes and to implement these effectively in responses to changing market demands and technological advancements (Teece 2012; Bauweraerts and Colot 2023; Mahajan et al. 2024). The performance of electronics companies is heavily driven by their capacity for innovation, both globally and in emerging markets such as Bangladesh (Octasylva et al. 2022; Bauweraerts and Colot 2023). Innovation has become a key driver in the competitive, fast-evolving digital landscape, allowing firms to survive and thrive (Bauweraerts and Colot 2023). The dynamic nature of technology and shifting customer preferences requires constant adaptation, making innovation capability crucial to maintaining and enhancing firm performance (Yang and Aumeboonsuke 2022; Bauweraerts and Colot 2023). By investing in R&D, adopting cutting-edge technologies, and fostering a culture of creativity, electronics companies can create distinctive value propositions that bolster their competitive advantage on the global stage (Parra-Requena et al. 2022; J. Sun et al. 2023; X. Huang 2023; Mahajan et al. 2024).

Moreover, innovation capability is central to firm sustainability and longevity in the fast-changing electronics companies, where continuous improvement is critical (Teece 2012; Bauweraerts and Colot 2023; Mahajan et al. 2024). The assessment of innovation capabilities often utilizes various metrics to evaluate electronics companies performance, with product, process, technological, organizational, and marketing innovations among the most frequently considered dimensions (Hossain and Azam 2023; Sindarov et al. 2023; Mahajan et al. 2024). Product innovation focuses on the creation of new products or services, while process innovation aims to enhance operational efficiency (Reimann, Carvalho, and Duarte 2021; Hossain and Azam 2023; Mahajan et al. 2024). Technological innovation involves the adoption of new technologies, while organizational innovation addresses changes in company structure and processes (Groza, Zmich, and Rajabi 2021; Çağlıyan, Attar, and Abdul-Kareem 2022; Mahajan et al. 2024). Marketing innovation, on the other hand, includes novel ways of engaging with and reaching consumers.

The relationship between innovation capability and firm performance is not always linear. While some empirical research shows a strong, positive link between innovation capability and firm performance, other studies suggest a more complex, nonlinear association, influenced by external factors such as market volatility and competition intensity (Teece 2012; Yousaf et al. 2020; Urban and Matela 2022; Mahajan et al. 2024; Nguyen, Nguyen, and Nguyen 2023). Product and process innovations, in particular, often enhance performance, but their effectiveness may be mitigated by industry-specific challenges (Mahajan et al. 2024). Many studies highlight the ability of innovation to amplify the benefits of technological advancements in firms, thus improving overall performance (Parra-Requena et al. 2022; J. Sun et al. 2023; Mahajan et al. 2024).

Understanding the intricate link between innovation capability and firm performance is critical for effective strategic planning and decision-making (Waqas and Tan 2023). Managers can utilize these insights to develop innovative strategies that align with their company's objectives and competitive environment (Y. Huang et al. 2022; Parra-Requena et al. 2022; J. Sun et al. 2023; Mahajan et al. 2024). Future research should focus on addressing methodological limitations and exploring the influence of emerging technologies, such as blockchain and artificial intelligence, on firm performance. It is also essential to study how cultural differences shape innovation strategies and outcomes (Yuliansyah et al. 2021; Parra-Requena et al. 2022; J. Sun et al. 2023). In Bangladesh, this relationship takes on special significance. As a developing nation with a rapidly growing digital economy, Bangladesh offers unique opportunities for innovation-driven firms. Companies can capitalize on increasing internet penetration and evolving consumer preferences to enhance their market positioning (Jun, Lee, and Park 2021; Ali 2022; Parra-Requena et al. 2022; J. Sun et al. 2023; Mahajan et al. 2024).

Moreover, blockchain technology known for its decentralized, transparent, and secure nature has perceived increasing applications in supply chain management, financial services, and other industries. However, blockchain adoption requires firms to invest in significant technological infrastructure, rethink their operational processes, and sometimes overcome regulatory hurdles (J. Sun et al. 2023; Mahajan et al. 2024). The ability to innovate enables firms to handle these challenges more effectively, making it easier for them to adopt cutting-edge technologies and to transition to blockchain-based systems (Yuliansyah et al. 2021; Parra-Requena et al. 2022). Therefore, firms that possess strong innovation capabilities are better positioned to adopt and benefit from blockchain technology. Based on this rationale, it is hypothesized that:

H1. *Innovation capability has a positive and significant impact on firm performance.*

H2. *Innovation capability has a positive and significant impact on adoption of blockchain technology.*

2.2 | Information-Sharing Capability and Adoption of Blockchain Technology

The connection between information-sharing capability and firm performance is critical, particularly in rapidly digitizing economies like Bangladesh where the reliance on technology and digital platforms for operations is growing swiftly (Phung et al. 2021; Sindarov et al. 2023). Information-sharing capability refers to a firm's ability to exchange data, insights, and knowledge both internally and externally. In today's highly competitive and digitally connected business environment, this capability becomes fundamental for firm success. Effective information sharing facilitates better decision-making, increases

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operational efficiency, enhances transparency, and creates synergies within supply chains and partnerships (Muhamed et al. 2023). In an increasingly data-driven world, organizations that excel in sharing information can better align their strategies, reduce inefficiencies, and innovate based on real-time market insights (Sindarov et al. 2023).

For firms operating in a digital-first environment, sharing data with stakeholders, customers, and other businesses within the value chain significantly impacts firm performance. The ability to strategically exchange information can lead to improved responsiveness, reduced operational inefficiencies, and better alignment with market demands. In both stable and rapidly changing markets, the implementation of interorganizational information-sharing strategies has shown a positive and significant effect on firm performance (Chuang and Lin 2015; Dhewanto et al. 2020). This is especially true in supply chains where a spectrum of information-sharing models from minimal to comprehensive transmission has been observed (Panahifar et al. 2018; Şahin and Topal 2019; Phung et al. 2021).

Researchers have extensively explored the impact of communication strategies, information sharing, and supplier development on firm performance. The ability of firms to efficiently exchange information within and between organizations strengthens supplier development efforts and supports firm growth (Panahifar et al. 2018; Şahin and Topal 2019; Phung et al. 2021). The study by Tajvidi and Karami (2021) significantly contributes to the understanding of how supply chain integration, information sharing, and overall supply chain architecture influence firm performance. This research underlines the importance of robust information-sharing capabilities in fostering better relationships between buyers and suppliers, subsequently boosting overall firm performance (Yu et al. 2019; Ho and Kuvaas 2020; Rajala and Hautala-Kankaanpää 2023).

Moreover, Hanelt et al. (2021) highlighted that aligning information technology (IT) systems and processes with supply chain needs enhances operational performance. Their study found that high-quality, integrated IT systems positively impact the depth and breadth of information sharing across supply chains. This, in turn, improves collaboration, reduces transaction costs, and increases operational efficiencies (Braojos, Benitez, and Llorens 2019; Rajala and Hautala-Kankaanpää 2023).

For Bangladeshi electronics companies navigating digital transformation, leveraging information sharing has proven beneficial for optimizing supply chains, improving customer satisfaction, and streamlining internal processes (Caputo et al. 2019; Kristoffersen et al. 2021; Shahzad et al. 2023). In the globalized business landscape, information sharing is a strategic tool that enables companies to collaborate, innovate, and swiftly adapt to dynamic markets and evolving customer preferences (Chang, Wang, and Arnett 2018; Corral de Zubielqui et al. 2019; Shahzad et al. 2023). Companies that excel in cross-border information exchange gain a competitive edge in international trade by enhancing marketing strategies, supply chain operations, and overall business efficiency (Chang, Wang, and Arnett 2018; Corral de Zubielqui et al. 2019; Shahzad et al. 2023; Yu et al. 2019; Rajala and Hautala-Kankaanpää 2023). Therefore, based on the growing importance of information sharing and its positive impacts on firm performance in both local and global contexts, it was hypothesized that

H3. *Information-sharing capability has a positive and significant impact on firm performance.*

On the other hand, we know that blockchain technology relies heavily on the accurate, transparent, and secure exchange of information across multiple parties (Chang, Wang, and Arnett 2018; Corral de Zubielqui et al. 2019). Companies that already have strong information-sharing capabilities are well positioned to adopt blockchain technology because they possess the necessary infrastructure to exchange, verify, and store data across various stakeholders (Shahzad et al. 2023). In addition, adoption of cutting-edge technologies like blockchain adoption often requires a collaborative, network-based approach, as it involves various entities sharing a distributed ledger (Shahzad et al. 2023). Companies with established information-sharing mechanisms are already familiar with the practices needed for efficient coordination, data integrity, and stakeholder involvement, making the transition to blockchain technology smoother (Rajala and Hautala-Kankaanpää 2023). For example, effective information-sharing processes enhance trust between parties, reduce data redundancies, and streamline workflows, all of which are critical for blockchain's successful implementation. In resource-constrained markets like Bangladesh, where firms often face challenges such as limited technological infrastructure and evolving regulatory environments, strong information-sharing capabilities can serve as a competitive advantage (Teece 2012). These capabilities allow businesses to leverage blockchain as a tool to enhance their existing data-sharing frameworks, improve supply chain transparency, and ensure secure transactions (Braojos, Benitez, and Llorens 2019; Rajala and Hautala-Kankaanpää 2023). By facilitating trust and collaboration among stakeholders, firms with advanced information-sharing systems are more likely to view blockchain adoption not only as feasible but also as an opportunity to further optimize their supply chains and build resilience. Thus, we hypothesized the following:

H4. *Information-sharing capability has a positive and significant impact on adoption of blockchain technology.*

2.3 | Technological Capability and Adoption of Blockchain Technology

In today's fast-changing business environment, technology plays a pivotal role in shaping organizational success and failure (Chae, Koh, and Park 2018; Heredia et al. 2022; Shah et al. 2024). As digital transformation accelerates, the ability of companies to leverage and optimize technological capabilities becomes a critical determinant of their competitiveness and overall market success (Kristoffersen et al. 2021; Shah et al. 2024). In this regard, Valdez-Juárez and Castillo-Vergara (2021) have highlighted the strong relationship between IT proficiency and firm performance, underlining how effective use of technology can lead to substantial gains in efficiency, innovation, and profitability.

The role of business IT capabilities in enhancing firm performance is well documented, with Kristoffersen et al. (2021)

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and Shah et al. (2024) evaluating IT infrastructure, human resources, and relationship management as crucial factors. Organizations with advanced IT capabilities are better positioned to innovate, streamline operations, and maintain agility in an ever-changing business landscape (Rajapathirana and Hui 2018; Ferreira, Coelho, and Moutinho 2020; Q. Chen, Wang, and Huang 2021). By categorizing companies' IT resources ranging from IT infrastructure to IT-related capabilities, researchers have developed models that demonstrate how these competencies create a competitive advantage (Y. Chen and Lin 2021). According to Camisón-Haba, Clemente-Almendros, and Gonzalez-Cruz (2019), companies that align their IT resources with industry-specific demands can maximize performance, illustrating how the RBV frames technological capability as a source of sustained competitive advantage.

Furthermore, Chege, Wang, and Suntu (2020) propose a conceptual model that connects performance management, process management, and customer management with IT-enabled information management capabilities. These insights underscore the multifaceted role of technological capability in supporting not only operational excellence but also strategic decision-making and market responsiveness (Kristoffersen et al. 2021; Shah et al. 2024).

Technological capability, as defined by investments in R&D, access to cutting-edge technologies, and the ability to integrate these technologies effectively, provides firms with the tools to innovate and adapt in competitive global markets (Jin, García, and Salomon 2019; Kristoffersen et al. 2021; Shah et al. 2024). Multinational corporations in particular leverage their vast technological resources to optimize supply chains, create innovative products, and expand market presence (Ferraris et al. 2019; Mikalef and Gupta 2021). In this context, companies that excel in digital capabilities ranging from online sales and digital marketing to the use of big data analytics are likely to outperform competitors and capture a larger share of the global market (Singh et al. 2019; Kristoffersen et al. 2021; Shah et al. 2024).

In emerging markets such as Bangladesh, the connection between technological capabilities and firm performance is evolving (Peerally, De Fuentes, and Figueiredo 2019; Bahta et al. 2021; Hoque et al. 2023; Shah et al. 2024). While technological advancements in the region have enabled firms to begin harnessing digital tools for growth, challenges such as limited infrastructure, skills shortages, and financial constraints continue to impede full digital adoption (Rahman et al. 2023). Nonetheless, government-led initiatives to promote digitalization, coupled with expanding internet access, are fostering an environment conducive to technological advancement (Bahta et al. 2021; Rahman et al. 2023; Shah et al. 2024). Companies that are able to overcome these barriers and capitalize on their technological capabilities are likely to experience significant improvements in performance and innovation potential. Thus, it was hypothesized that

H5. *Technological capability has a positive and significant impact on adoption of firm performance.*

On the other hand, business technological capability can also play a critical role in the adoption of emerging technologies, such as blockchain. Companies with the ability to integrate and utilize advanced technologies are more likely to recognize the benefits of blockchain, including efficiency in transactions and data management (Hoque et al. 2023; Shah et al. 2024). As blockchain requires sophisticated digital infrastructure and the capacity to handle complex data exchanges, companies with robust technological capabilities are better positioned to adopt and implement blockchain technology effectively, thereby gaining a competitive edge in markets that prioritize secure, decentralized data systems (Peerally, De Fuentes, and Figueiredo 2019; Bahta et al. 2021). By focusing on the strategic deployment of technology, electronics companies can drive innovation and ensure their operations are aligned with cutting-edge advancements such as blockchain, which can enhance transparency, reduce fraud, and improve interorganizational collaboration (Kristoffersen et al. 2021; Shah et al. 2024). In the electronics sector, technological capability plays an even more pivotal role, as firms in this industry are often at the forefront of digital innovation. Companies that invest in developing their technological capabilities are better positioned to adopt blockchain as a strategic initiative to align with global trends in secure and transparent data systems. This capability not only enables firms to overcome the technical complexities associated with blockchain but also ensures that they can integrate it into their existing operations to drive innovation, streamline workflows, and enhance overall competitiveness (Teece 2012; Shah et al. 2024).

H6. *Technological capability has a positive and significant impact on adoption of blockchain technology.*

2.4 | Adoption of Blockchain Technology and Firm Performance

The adoption of blockchain technology by companies can significantly influence a variety of performance metrics. The adoption of blockchain technology can enhance operational efficiency by streamlining transactions, reducing paperwork, minimizing intermediaries, and automating processes (Fernando, Rozuar, and Mergeresa 2021; Deng, Gong, and Wang 2024). These improvements can lead to lower costs, faster settlement times, and better utilization of resources, ultimately enhancing firm performance. The inherent cryptographic features of blockchain provide a high level of security and integrity for online transactions and data storage (Kamble, Gunasekaran, and Arha 2019; Chaudhuri, Chatterjee, and Vrontis 2024). By mitigating risks associated with fraud, manipulation, and data breaches, the adoption of blockchain technology can boost stakeholder confidence, thereby enhancing overall firm performance (Farnoush et al. 2022; Chaudhuri, Chatterjee, and Vrontis 2024). Additionally, blockchain has the potential to revolutionize supply chain management by providing accountability, traceability, and end-to-end visibility. Improved inventory management, expedited dispute resolution, reduced counterfeit products, and enhanced regulatory compliance collectively contribute to better firm performance (Bag et al. 2023; Deng, Gong, and Wang 2024). Moreover, by offering transparent and immutable records of transactions and interactions, blockchain adoption can bolster consumer trust and loyalty. Consequently, higher customer satisfaction rates are likely to lead to increased usage of products and services, further enhancing firm

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performance and reinforcing trust in the integrity of goods and transactions (Kamble, Gunasekaran, and Arha 2019; Bag et al. 2023; Chaudhuri, Chatterjee, and Vrontis 2024). Thus, it was hypothesized that

H7. *Adoption of blockchain technology has a positive and significant impact on firm performance.*

2.5 | Innovation Capability, Adoption of Blockchain Technology, and Firm Performance

Innovation capability has emerged as a critical determinant for firms navigating the complexities of the digital age (Octasylva et al. 2022; Bauweraerts and Colot 2023). Companies that prioritize innovation and spend more on R&D projects are not only more adept at recognizing emerging technologies but they are also more likely to leverage the transformative potential of blockchain technology. The strategic adoption of blockchain is fuelled by a robust culture of innovation aimed at enhancing security, efficiency, and trustworthiness across organizational processes (Parra-Requena et al. 2022; J. Sun et al. 2023; X. Huang 2023).

In the rapidly evolving landscape of the global electronics industry, the benefits of blockchain adoption are increasingly evident. Companies that successfully implement blockchain technology can streamline their operations, reduce costs, and boost their competitiveness in a fast-growing market. By integrating blockchain into their business models, these firms enhance their ability to respond to customer demands, ensure data integrity, and facilitate smoother transactions (Bauweraerts and Colot 2023).

In emerging economies like Bangladesh, the dynamics of innovation play a pivotal role in fostering blockchain technology adoption. As the country experiences economic growth, companies that cultivate a culture of innovation are better positioned to perceive blockchain as a viable solution to prevalent challenges related to data security and transparency (Mahajan et al. 2024). This is particularly relevant in sectors such as finance, logistics, and e-commerce, where the implementation of blockchain can significantly enhance operational efficiency and consumer trust (Fernando, Rozuar, and Mergeresa 2021; Sciarelli et al. 2022; Hossain and Azam 2023; Sindarov et al. 2023; Mahajan et al. 2024; Deng, Gong, and Wang 2024). Therefore, it is hypothesized that the adoption of blockchain technology plays a mediating role in the relationship between innovation capability and firm performance, providing a strategic pathway for firms to leverage their innovative strengths for enhanced competitive advantage.

H8. *Adoption of blockchain technology mediates the association between innovation capability and firm performance.*

2.6 | Information-Sharing Capability, Adoption of Blockchain Technology, and Firm Performance

Global information sharing has emerged as a cornerstone of the collaboration ecosystem, fostering partnerships and enhancing organizational resilience (Phung et al. 2021; Sindarov et al. 2023; Muhamed et al. 2023). A strong information-sharing capability not only improves internal business processes and facilitates high quality decision-making but also enhances external collaborations (Chuang and Lin 2015; Dhewanto et al. 2020). Companies that prioritize transparent and efficient data sharing are better positioned to recognize and leverage the transformative potential of new technology like blockchain. This technology represents a strategic response to contemporary challenges, offering improved data security, operational efficiency, and heightened trust among stakeholders (Chuang and Lin 2015; Dhewanto et al. 2020; Panahifar et al. 2018; Şahin and Topal 2019; Phung et al. 2021).

On the other hand, the adoption of blockchain technology allows firms to address critical issues such as data integrity and fraud risk while enhancing confidence in their operational processes. By enabling secure, real-time access to shared information, blockchain facilitates informed decision-making and improves responsiveness to market changes (Chuang and Lin 2015; Dhewanto et al. 2020). Consequently, companies can experience substantial gains in productivity through increased efficiency, reduced risk of fraud, and greater trust among partners and customers. Organizations that embrace information sharing can effectively leverage blockchain technology to create competitive advantages, streamline operations, and enhance overall performance (Sindarov et al. 2023; Muhamed et al. 2023).

By fostering a culture of transparency and open communication, organizations can drive the adoption of blockchain solutions, which in turn translates into improved firm performance. Thus, it is hypothesized that the adoption of blockchain technology serves as a mediating factor in the relationship between information-sharing capability and firm performance, positioning companies to fully capitalize on their information-sharing practices for enhanced operational success (Şahin and Topal 2019; Phung et al. 2021).

H9. *Adoption of blockchain technology mediates the association between information-sharing capability and firm performance.*

2.7 | Technological Capability, Adoption of Blockchain Technology, and Firm Performance

Technological capabilities serve as a critical foundation that empowers firms to explore, adopt, and integrate new technologies effectively (Chae, Koh, and Park 2018; Heredia et al. 2022; Shah et al. 2024). In the global marketplace, organizations equipped with advanced technological capabilities are strategically positioned to recognize and leverage the potential benefits of blockchain technology (Kristoffersen et al. 2021; Shah et al. 2024). The adoption of blockchain is not merely a technical upgrade; it is a strategic imperative for firms aiming to enhance the security, transparency, and efficiency of their operations. These improvements can significantly contribute to better overall firm performance (Rajapathirana and Hui 2018; Ferreira, Coelho, and Moutinho 2020; Q. Chen, Wang, and Huang 2021; Shah et al. 2024).

In Bangladesh, where the digital landscape is undergoing rapid evolution, the relationship between technological capability and

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blockchain technology adoption assumes a transformative significance (Kamble, Gunasekaran, and Arha 2019; Chaudhuri, Chatterjee, and Vrontis 2024). As electronics companies enhance their technological capabilities, they increasingly recognize the transformative potential of blockchain to address pressing challenges, including data security vulnerabilities and the need for transaction transparency. This recognition leads to a higher likelihood of adopting blockchain solutions, which when effectively implemented, can markedly enhance firm performance (Chuang and Lin 2015; Dhewanto et al. 2020).

The impact of blockchain technology adoption is particularly pronounced in sectors such as finance, supply chain management, and e-commerce, where operational transparency and data integrity are paramount (Ferraris et al. 2019; Mikalef and Gupta 2021; Fernando, Rozuar, and Mergeresa 2021; Deng, Gong, and Wang 2024). In these industries, firms that adopt blockchain technology not only improve their internal efficiencies but also enhance their competitive positioning in the market. The ability to harness advanced technological capabilities allows firms to integrate blockchain solutions, which subsequently leads to improved operational effectiveness and performance outcomes (Mikalef and Gupta 2021; Fernando, Rozuar, and Mergeresa 2021). Thus, it is proposed that the adoption of blockchain technology mediates the association between technological capability and firm performance, facilitating a pathway for firms to leverage their technological investments for competitive advantage (Figure 1).

H10. *Adoption of blockchain technology mediates the association between technological capability and firm performance.*

3 | Methodology of the Study

3.1 | Study Context

Small- and medium-sized enterprises (SMEs) constitute a significant portion, approximately 90%, of businesses worldwide (Costa Melo et al. 2023). The impact of digital transformation differs for electronics companies compared to larger corporations due to their greater adaptability and nimbleness in response to changing conditions. Nonetheless, electronics companies face challenges due to their limited resources and specialization capacities. Therefore, it is crucial to assess the performance of electronics companies from various angles, particularly in light of emerging technologies like blockchain technology (Polas and Raju 2021). Our research focuses on electronics companies within the Bangladeshi electronics manufacturing sector, which contributes around one-third of global energy consumption and CO_2 emissions. However, there has been a recent emphasis on adopting a *green mindset* (Polas and Raju 2021) in this industry. Electronics manufacturers in Bangladesh produce a diverse array of products, catering to both domestic and international markets. In the realm of consumer electronics, the sector offers LED and Smart TVs, refrigerators, freezers, air conditioners, washing machines, microwave ovens, and mobile phones with accessories. Home appliances such as fans, rice cookers, blenders, grinders, irons, and water purifiers are also widely manufactured. The industrial electronics segment includes LED lighting solutions, electrical cables and wires, and batteries (e.g., lithium-ion and lead–acid), as well as generators and inverters. In addition, the sector produces computer and IT products, including desktop computers, laptops, and computer peripherals like keyboards, mice, and monitors, alongside networking equipment. Manufacturers also produce essential components and accessories such as printed circuit boards (PCBs), switches, sockets, plugs, transformers, and adapters. Furthermore, with the growing focus on renewable energy, the industry has expanded to include solar panels, solar inverters, and charge controllers, supporting the country's push towards sustainable energy solutions.

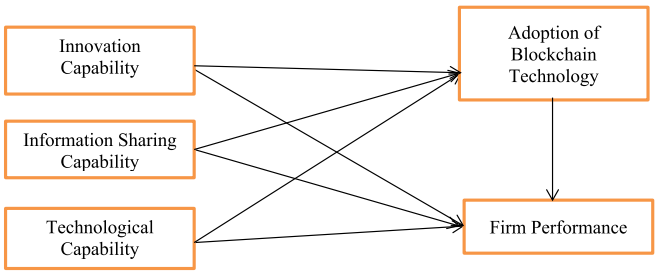

FIGURE 1 | The framework of the study (developed by the authors).

It worth to know that the supply chain process in electronics manufacturing is a complex and interconnected system that transforms raw materials into finished products and ensures their efficient delivery to customers. It begins with the sourcing of raw materials and components, such as semiconductors, PCBs, microprocessors, plastics, metals, and rare earth elements, often procured from specialized global suppliers. These materials are transported to manufacturing facilities through well-organized inbound logistics, involving inventory management and supplier coordination. The production phase involves transforming these inputs into finished goods through processes like molding, soldering, assembly, and testing, often utilizing advanced technologies such as robotics and automation. Quality control is a critical step, ensuring products meet stringent industry standards and customer expectations. Once finalized, finished products are transported to distribution centers or directly to retailers and customers via outbound logistics, which includes warehousing, packaging, and transportation management. Distribution occurs through various channels, including wholesalers, retailers, and e-commerce platforms. After-sales services, such as repairs, warranties, and returns, are managed through reverse logistics, which may include recycling or refurbishing to minimize environmental impact. Additionally, electronics manufacturers increasingly rely on technologies like blockchain, IoT, and ERP systems to enhance transparency, traceability, and efficiency across the supply chain, enabling real-time tracking and better collaboration among stakeholders. This dynamic process requires seamless coordination to navigate challenges such as fluctuating demand, resource constraints, and regulatory compliance.

3.2 | Research Design

In this study, the positivist research philosophy was adopted to empirically investigate the impact of organizational

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capabilities on firm performance and the mediating role of adoption of blockchain technology in electronics companies. The positivist approach is suitable due to its focus on objective, measurable phenomena, allowing for the formulation and testing of hypotheses using quantitative methods like structural equation modelling (SEM; Polas and Raju 2021). This philosophy brings into line the study's goal to develop empirical knowledge from observable data, ensuring the reliability and validity of findings and supporting generalizations to similar contexts within the electronics sector in Dhaka, Bangladesh (Hair et al. 2014). The choice of the Bangladeshi sample for this study is appealing due to the government's implementation of various sustainability policies and the establishment of tailored programs to promote production and engagement in climate-conscious emerging technologies within the industrial sectors (Polas and Raju 2021).

3.3 | Questionnaire Design

The final survey instrument encompassed 27 items (see Appendix A), addressing five variables aimed at tackling pertinent issues. Prior to finalizing the sample, a pretest was conducted to ensure the questionnaire's content validity. For the development of the survey instrument, we integrated four items sourced from Bauweraerts and Colot (2023) to assess innovation capability. Additionally, four information-sharing capability items (see Appendix A) were adopted from Sindarov et al. (2023). From Shah et al. (2024), four technological capability items were selected. The adoption of blockchain technology was measured using five items taken from Polas et al. (2022). Furthermore, to measure firm performance, five items from S. Kim, Lee, and Lim (2021) were employed. A Likert scale, encompassing five response categories ranging from *strongly disagree* (1) to *strongly agree* (5), was employed for assessment purposes.

3.4 | Sampling and Data Collection

In the context of Bangladesh, we carried out a cross-sectional study to explore the relationship between organizational capabilities and the adoption of blockchain technology as practices aimed at enhancing firm sustainability and performance. The data were collected from the operations and supply chain managers from electronics companies. We requested and received the names and contact information of supply chain managers from the HR department. From this list, part-time and seasonal managers were removed to maintain sample consistency. The stratified random sampling was adopted following the cross-sectional study during the months of September and October, 2023.

Initially, the questionnaire was dispatched to the operations/supply chain departments of 251 eligible electronics companies listed by and collected from the Dhaka Chamber of Commerce and Industry (DCCI). The data were collected from privately owned electronics companies at Dhaka division (Manikganj and Gazipur) in Bangladesh. Manikganj and Gazipur are located to the west and the north of Dhaka, Bangladesh. We primarily approached these managers during their lunch and dinner breaks to invite them to participate in our research. We assured them that their responses would remain confidential and that only the principal investigator would have access to their answers. Out of the initial list (251 managers, 1 manager from each firm), 37 managers declined to participate in the study, five were unreachable due to different reasons, and 16 provided incomplete answers. Finally, 198 organizations' supply chain managers returned completed surveys. Based on a sample of survey respondents, our survey had a 78.88% response rate.

Similar to other studies in this context, we employed the translation and back-translation method to convert the English survey into Bengali. The Bengali version was pilot-tested with five workers in the same industrial sectors. Data collection was facilitated by three trained local research assistants. We designed the surveys to be concise to encourage participation. Participation in the survey was voluntary, and informed consent was obtained prior to distribution. The survey was designed to take approximately 6–8 min to complete, and no personal questions were included to ensure participants' privacy. This technique involves dividing the population into different strata or groups based on certain characteristics (e.g., firm size, years of operation, or technological capability) and then randomly selecting samples from each stratum.

We used the time-trend extrapolation test recommended by Armstrong and Overton (1977), a widely used technique in academics, business, psychology, and other fields to address potential nonresponse bias in our work. In this test, the traits of the first 25% of respondents are compared to those of the subsequent respondents. We can identify any possible biases resulting from nonresponses by closely examining this comparison. An effective way to assess the influence of nonresponse bias on our results is to use the time-trend extrapolation test. With this approach, we can analyze the differences between early and late responders in a methodical way, which helps us spot any inconsistencies that might affect how the data are interpreted. Also, we strengthen the validity and reliability of our study's conclusions by utilizing a method that is well accepted and supported in other domains. The time-trend extrapolation test, therefore, ensures the validity of our results by adding to the scientific rigor and integrity of our study.

Figure 2 is provided as a reference for the methodological approach that we used to direct our study efforts.

To provide a comprehensive understanding of the study participants, Table 1 presents the demographic profile of our respondents. The sample consisted of 198 individuals, whose backgrounds varied in gender, working experience, firm size, functional area, and the primary business goal of their firms within the supply chain.

The sample was predominantly male (77.78%), with females representing 22.22% of the respondents. Regarding firm size (based on employee count), 61.62% of the respondents worked in firms with 50–200 employees, 21.72% in firms with 201–400 employees, and 16.67% in firms with 401–500 employees. Most respondents' firms served as first-tier suppliers to major firms (64.14%), with 22.22% as second-tier suppliers, 10.61%

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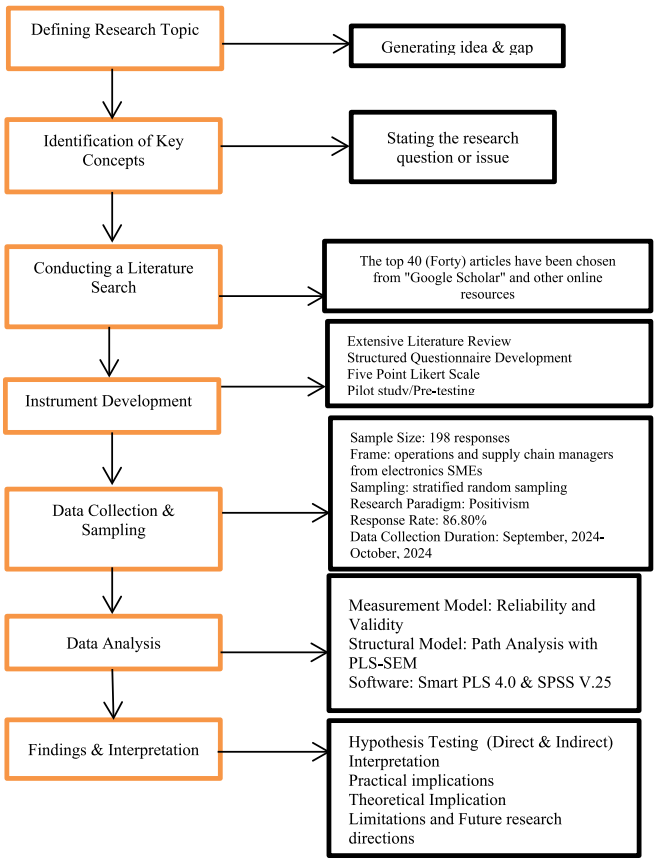

FIGURE 2 | The research process (developed by authors).

as suppliers to government agencies, and 3.03% categorized as other. Participants reported a range of working experience, with 22.22% having less than 5 years, 44.44% between 5 and 9 years, 13.64% between 9 and 11 years, 7.07% between 12 and 14 years, and 2.53% with over 15 years of experience. The participants held various roles, including production and operations management (11.62%), warehouse/inventory (9.09%), sales and distribution (16.67%), retail (27.27%), customer relationship (6.06%), and suppliers and distributors (29.29%). Including these demographic characteristics of the respondents enhances the transparency of our sample and offers context to better understand the relevance and applicability of our findings.

TABLE 1 | Respondent's demographic profile.

Characteristics	Frequency	Percentage
Gender		
Male	154	77.78
Female	44	22.22
Firm size (# of employees)		
50–200	122	61.62
201–400	43	21.72
401–500	33	16.67
Firm's primary business goal in supply chain		
First-tier supplier to major firms	127	64.14
Second-tier supplier	44	22.22
Supplier to government	21	10.61
Others	6	3.03
Working experience		
Less than 5 years	44	22.22
5–9 years	88	44.44
9–11 years	27	13.64
12–14 years	14	7.07
15 years or above	5	2.53
Functional area		
Production and operations management	23	11.62
Warehouse/inventory	18	9.09
Sales and distribution	33	16.67
Retail	54	27.27
Customer relationship	12	6.06
Suppliers and distributors	58	29.29
Total-198		

Source: Developed on SPSS V.25.

3.5 | Data Analysis

We used SEM to validate our study's research model. We confirmed the data's practicality, validity, and reliability by using the software packages Smart PLS 4.0 and SPSS V.25, with the goal of reaching significant statistical conclusions. We surpassed the required sample size for SEM analysis by adhering to the minimal sample-to-item ratio recommended by Hair et al. (2016), confirming the appropriateness of partial least squares SEM (PLS-SEM) for predictive purposes (Hair et al. 2016). The sample-to-item ratio is a commonly used guideline to ensure sufficient sample size. According to Hair et al. (2016), a minimum sample-to-item ratio of 5:1 is recommended, meaning at least five respondents per measurement item are needed to achieve reliable model estimation. In our study, we have 22 items measuring the main variables, resulting in a minimum required sample size of 110 respondents (22 items × 5 respondents per item). To ensure robustness and enhance predictive validity, we collected data from 198 respondents, resulting in a sample-to-item ratio of approximately 9:1 (198 respondents/22 measurement items), which surpasses the minimum requirement. This confirms the appropriateness of using PLS-SEM for predictive purposes (Hair et al. 2016). The selection of SEM as the solution method is justified by the research objectives of the study. SEM is the most appropriate method for this research because it allows

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for the simultaneous testing of multiple relationships, both direct and mediating, within our proposed conceptual framework (Hair et al. 2016).

3.6 | Mediation Analysis

We employed a two-step procedure proposed by Hair et al. (2016) to verify whether the adoption of blockchain technology holds any mediating influence. Additionally, we account for the indirect effects of innovation capability (IC), information-sharing capability (ISC), and technological capability (TC) on firm performance (FP) through adoption of blockchain technology (ABT) in the first step. We observe statistically significant results signalling an indirect effect for the following relationships: IC → ABT → FP; ISC → ABT → FP; and I → ADT → FP. The exact particulars of these results are viewable in Table 7. Next, we focus on the direct effect of IC, ISC, and TC on FP keeping intact the role of the mediator (ABT). Significant positive effects emerge for H8, H9, and H10. H8 provides the full mediation effect; H9 and H10 provide partial mediations.

4 | Findings

4.1 | Measurement, Validity, and Reliability

Table 2 evaluates the conceptual framework's assessment, validity, and reliability. All constructs meet the average variance extracted (AVE) benchmark of 0.50, confirming strong validity, while composite reliability (CR) and Cronbach's alpha exceed acceptable thresholds (Polas et al. 2022). Indicator loadings above 0.60 further demonstrate cutoff psychometric properties (Vinzi, Trinchera, and Amato 2010). The model shows that adoption of blockchain technology significantly impacts exogenous factors (44.7%), and firm performance has a significant influence (58%). The normed fit index (NFI) approaching unity and SRMR score of 0.08 indicate a good fit between the model and empirical data (Polas et al. 2022).

Table 3 highlights the model's strong predictive relevance, with Q^2 values above zero confirming its relevance (Chin 1998). The f^2 metric shows that the adoption of blockchain technology has a large effect, while innovation capability, information-sharing capability, and technological capability have smaller impacts on

TABLE 2 | Measurement of model assessment.

Constructs	Items	Loading	AVE	CR	Alpha	R^2	NFI	SRMR
Innovation capability	IC1	0.757						
	IC2	0.857	0.656	0.831	0.825			
	IC3	0.845						
	IC4	0.774						
Information-sharing capability	ISC1	0.660						
	ISC2	0.737	0.578	0.779	0.757			
	ISC3	0.814						
	ISC4	0.814						
Technological capability	TC1	0.713						
	TC2	0.731	0.504	0.675	0.675			
	TC3	0.69						
	TC4	0.706						
Adoption of blockchain technology	ABT1	0.754						
	ABT2	0.809						
	ABT3	0.839	0.619	0.849	0.845	0.447		
	ABT4	0.805						
	ABT5	0.720						
Firm performance	FP1	0.776						
	FP2	0.78						
	FP3	0.832	0.643	0.862	0.861	0.580	0.901	0.071
	FP4	0.785						
	FP5	0.834						

Abbreviations: AVE: average variance extracted; CR: composite reliability; NFI: normed fit index; SRMR: standardized root mean square residual.

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TABLE 3 | Values of the Stone–Geisser indicator (Q^2) and Cohen's indicator (f^2) of the model in the SEM.

Variables	Q^2 predict	Adoption of blockchain technology (f^2)	Firm performance (f^2)
Firm performance	0.394		
Information-sharing capability		0.140	0.034
Innovation capability		0.137	0.002
Adoption of blockchain technology	0.429		0.391
Technological capability		0.07	0.046

Note: Large effect > 0.34; medium effect > 0.14; small effect > 0.01 (Cohen 1988).

firm performance. Information-sharing capability has a moderate effect on the adoption of blockchain technology, while innovation capability and technological capability show modest effects. Overall, the model demonstrates commendable fitness and precision in predicting outcomes, affirming its empirical validity and relevance in the studied context.

4.2 | Discriminant Validity

4.2.1 | Fornell–Larcker Criterion Analysis

Table 4 presents the estimations delineating the associations between the square roots of AVE and the latent variables manifested along the principal diagonal of the SEM. The bolded entries denote the square root of AVE, which surpasses the intervariable correlations, falling within the range of 0.710 to 0.810. As per Fornell and Larcker's (1981) established criteria, this outcome signifies a commendable discriminant validity appraisal.

4.2.2 | Heterotrait–Monotrait (HTMT) Analysis

Table 5 delineates the analysis of HTMT to assess discriminant validity. For validation purposes, HTMT values ought to remain below 0.85. Meeting this criterion, indicative of discriminant validity confirmation, is crucial as per the guidelines established by Henseler et al. (2005).

4.3 | Structural Model Assessment

The structural model assessment is crucial for validation. Using bootstrapping with 4999 resamples, t-values and R^2 were calculated. All items showed strong outer loadings above acceptable thresholds, and most variables had robust directional coefficients, confirming their strength within the model.

TABLE 4 | Fornell–Larcker criterion analysis for discriminant validity.

		1	2	3	4	5
1	Firm performance	**0.802**				
2	Information-sharing capability	0.529	**0.760**			
3	Innovation capability	0.432	0.338	**0.810**		
4	Adoption of blockchain technology	0.728	0.534	0.511	**0.786**	
5	Technological capability	0.531	0.468	0.409	0.509	**0.710**

Note: The diagonal are the square root of the AVE (in bold) of the latent variables and indicates the highest in any column or raw.

TABLE 5 | The heterotrait–monotrait (HTMT) analysis for discriminant validity.

		1	2	3	4	5
1	Firm performance					
2	Information-sharing capability	0.646				
3	Innovation capability	0.506	0.424			
4	Adoption of blockchain technology	0.848	0.649	0.611		
5	Technological capability	0.683	0.638	0.521	0.663	

Note: Discriminant validity exists if the HTMT < 0.85 (Henseler et al. 2005).

4.4 | Hypotheses Testing (Direct and Indirect Relationships)

The outcomes pertaining to the direct and indirect impact propositions are delineated in Table 6. The utilization of bootstrapping facilitated the evaluation of statistical t-values. Smart PLS 4.0, chosen for its suitability in social science research (Bickel 2012), was employed at a 95% confidence level. The first hypothesis posited a positive and significant association between innovation capability and firm performance ($\beta = 0.033$, $t = 0.664$; $p > 05$, refer to Table 6). The second hypothesis posited a positive and significant association between innovation capability and the adoption of blockchain technology. The results affirm this notion, indicating a significant and positive association between innovation capability and the adoption of blockchain technology ($\beta = 0.307$, $t = 5.689$; $p < 0.05$, refer to Table 6).

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TABLE 6 | Results of direct effect hypotheses.

Hypotheses	Relationships	Std beta	2.5% CI	97.5% CI	*t*-value	*p*-value	VIF	Decision
H1	Innovation capability → firm performance	0.033	−0.065	0.131	0.664	0.507	1.413	Rejected
H2	Innovation capability → ABT	0.307	0.203	0.411	5.689	0.000	1.242	Supported
H3	Information-sharing capability → firm performance	0.147	0.047	0.254	2.755	0.006	1.509	Supported
H4	Information-sharing capability → ABT	0.320	0.209	0.420	5.906	0.000	1.324	Supported
H5	Technological capability → firm performance	0.171	0.084	0.262	3.713	0.000	1.507	Supported
H6	Technological capability → ABT	0.234	0.129	0.332	4.556	0.000	1.408	Supported
H7	ABT → Firm Performance	0.545	0.418	0.652	9.105	0.000	1.809	Supported

Abbreviation: ABT: adoption of blockchain technology.

TABLE 7 | Results of indirect effect hypotheses.

Hypotheses	Relationships	Std beta	2.5% CI	97.5% CI	*t*-value	*p*-value	Decision
H8	Innovation capability → ABT → firm performance	0.167	0.103	0.244	4.673	0.000	Supported
H9	Information-sharing capability → ABT → firm performance	0.175	0.112	0.246	5.087	0.000	Supported
H10	Technological capability → ABT → firm performance	0.128	0.073	0.191	4.241	0.000	Supported

Abbreviation: ABT: adoption of blockchain technology.

Proceeding to the third hypothesis, it was posited that information-sharing capability has a positive and statistically significant association with firm performance. The empirical findings affirm this proposition, demonstrating a positive and significant association between information-sharing capability and firm performance ($\beta = 0.147$, $t = 2.755$; $p < 0.05$, see Table 6). Moving on to the fourth hypothesis, it was proposed that the information-sharing capability is positively and significantly associated with adoption of blockchain technology. This hypothesis is supported by the data, which show a positive and significant association between information-sharing capability and the adoption of blockchain technology ($\beta = 0.320$, $t = 5.906$; $p < 0.05$; see Table 6).

The fifth hypothesis proposed a positive and statistically significant association between technological capability and firm performance, which is consistent with the previous hypotheses. This claim is supported by an empirical study, which shows a significant and positive association between technological capability and firm performance ($\beta = 0.171$, $t = 3.713$; $p < 0.05$, see Table 6).

Following in the footsteps of other hypotheses, the sixth hypothesis proposed a positive and statistically significant association between technological capability and adoption of blockchain technology. The claim is supported by an empirical analysis, which indicates a significant and positive association between technological capability and the adoption blockchain technology ($\beta = 0.234$, $t = 4.556$; $p < 0.05$, see Table 6). Consistent with the patterns shown in previous hypotheses, the seventh hypothesis proposed a positive and statistically significant association between the adoption of blockchain technology and firm performance. Research supports this claim, revealing a positive and significant association between the adoption of blockchain technology and firm performance ($\beta = 0.545$, $t = 9.105$; $p < 0.05$; see Table 6).

Table 7 shows the results of indirect effect hypotheses. The degree to which a third variable referred to as a mediator clarifies the connection between two other variables is investigated through mediation analysis. The eighth hypothesis, which is consistent with this, suggests that the relationship between innovation capability and firm performance is mediated by the adoption of blockchain technology. Based on our analysis, as shown in Table 7, it can be concluded that the adoption of blockchain technology mediates the association between innovation capability and firm performance ($\beta = 0.167$, $t = 4.673$; $p < 0.05$).

Regarding the link between information-sharing capability and firm performance, the ninth hypothesis proposes that the adoption of blockchain technology acts as a mediator. We found, after analysis, that the adoption of blockchain technology does in fact mediate the relationship between information-sharing capability and firm performance ($\beta = 0.175$, $t = 5.087$; $p < 0.05$), shown in Table 7.

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The tenth hypothesis suggests that the link between technological capability and firm performance is mediated by the adoption of blockchain technology. As shown in Table 7, our study reveals that, in practice, the adoption of blockchain technology acts as a mediator in the relationship between technological capability and firm performance ($\beta = 0.128$, $t = 4.241$; $p < 0.05$).

5 | Discussion

Our study of electronics companies in Bangladesh aimed to examine organizational antecedents and consequences of blockchain technology adoption as sustainable supply chain practices. Our first hypothesis revealed no direct and significant impact of firms' innovation capabilities on performance. This finding contrasts with previous research (Octasylva et al. 2022; Bauweraerts and Colot 2023; Mahajan et al. 2024), which suggested a positive correlation between innovation capabilities and firm performance. This discrepancy highlights the need for further exploration of contextual factors that may influence these relationships, particularly in emerging markets like Bangladesh. Furthermore, our study found empirical support for a positive and significant impact of firms' innovation capability on blockchain technology adoption. This finding aligns with the results from Parra-Requena et al. (2022), Hossain and Azam (2023), and Sindarov et al. (2023). Consequently, an increase in the innovation capability in electronics companies triggers corresponding advancements in blockchain technology adoption, emphasizing the importance of fostering innovation to drive technological integration in supply chain processes.

In addition, our research confirms the positive and significant impact of information-sharing capability on firm performance, aligning with the findings of Şahin and Topal (2019), Phung et al. (2021), and Sindarov et al. (2023). These insights suggest that firms aiming to enhance performance should prioritize strengthening information-sharing practices, as effective communication within the supply chain plays a crucial role in driving organizational success. In this research, we also found that organizational information-sharing capability has a positive and significant impact on the adoption of blockchain technology in the supply chain process. This finding aligns with previous research by Braojos, Benitez, and Llorens (2019) and Rajala and Hautala-Kankaanpää (2023), further validating our results. The implications of this finding suggest that improving information-sharing practices within organizations not only enhances communication but also facilitates the adoption of advanced technologies like blockchain. By fostering a collaborative environment where knowledge flows freely, firms can more effectively integrate innovative solutions, ultimately driving efficiency and competitiveness in their supply chains.

It is worth mentioning that our research confirmed that technological capability positively contributes to firm performance, which is consistent with previous findings by Chae, Koh, and Park (2018), Heredia et al. (2022), and Shah et al. 2024. This reinforces the understanding that strengthening technological capabilities is essential for enhancing business outcomes, allowing firms to leverage innovations and improve operational efficiency. To further strengthen our conclusions, our findings also reveal that technological capability is crucial for the adoption of blockchain technology in supply chain processes, aligning with previous research by Jin, García, and Salomon (2019), Kristoffersen et al. (2021), and Shah et al. (2024). Firms with advanced technological capabilities are better positioned to understand the potential benefits of blockchain technology and assess its feasibility within their operations. Moreover, these firms are equipped to overcome barriers to adoption, such as the need for appropriate infrastructure and skilled personnel. Additionally, companies with strong technological capabilities tend to be more proactive in seeking innovative solutions to maintain a competitive edge, which can drive them to adopt emerging technologies like blockchain.

Our findings showed that the adoption of blockchain technology has a positive and significant impact on firm performance, consistent with previous studies by Bag et al. (2023) and Wang, Wong, and Boon-itt (2020). Several factors support this relationship. First, blockchain adoption reflects a forward-thinking approach to leveraging cutting-edge technologies, which can enhance operational efficiency, transparency, and security (Kristoffersen et al. 2021). These improvements often lead to increased customer satisfaction, productivity, and cost savings, all of which positively influence firm performance (Octasylva et al. 2022; Bauweraerts and Colot 2023). Additionally, businesses that adopt blockchain may attract partnerships and investments from stakeholders who value technological innovation, further boosting performance. Moreover, blockchain technology can facilitate the creation of new revenue streams and business models, contributing to overall firm growth. Hence, our data confirm a positive correlation between blockchain adoption and firm performance, reinforcing the strategic value of investing in such technologies (Mahajan et al. 2024).

We found empirical support for the role of blockchain technology as a mediator in the relationship between innovation capability and firm performance. The adoption of blockchain technology as an environmentally friendly practice exemplifies how dynamic capabilities—such as innovation capability—allow firms to integrate new, sustainable solutions into their operations (Shah et al. 2024). Blockchain can streamline supply chain processes, reduce waste, and enhance transparency, thereby promoting environmental sustainability (Teece 2012; Bauweraerts and Colot 2023). By leveraging this technology, firms can not only improve operational efficiency but also align their practices with sustainability goals, creating both environmental and economic value (Jin, García, and Salomon 2019). Through the lens of dynamic capabilities, innovation capability facilitates the identification and integration of blockchain as a strategic tool that enables firms to capture opportunities related to environmental responsibility (Teece 2012; Hossain and Azam 2023). This adoption is particularly significant as firms navigate an increasingly environmentally conscious market, where sustainability is becoming a critical factor in competitive advantage (Bauweraerts and Colot 2023). By emphasizing innovation capability as a dynamic capability that supports the adoption of blockchain, our study illustrates how firms can use such technologies to not only improve performance but also position themselves as leaders in environmental sustainability (Sindarov et al. 2023).

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We also found that the adoption of blockchain technology mediates the relationship between information-sharing capability and firm performance. Effective information sharing within a company is not just a passive exchange of data but a dynamic capability that enhances coordination, fosters creativity, and encourages collaboration (Teece 2012; Ahl et al. 2020). As electronic companies build stronger communication networks, they can more efficiently align their internal resources and expertise, which, in turn, enhances both operational performance and strategic decision-making (Barney 1991; Ducas and Wilner 2017). This reconfiguration of resources through information sharing enables electronic firms to dynamically respond to environmental shifts, a core aspect of RBV's extension into dynamic capabilities (Barney 1991; Eslami et al. 2024). By adopting blockchain, firms in the electronics sector are able to mobilize their dynamic capabilities—such as information-sharing and technological adaptation—allowing them to optimize their resource use, improve operational transparency, and respond more effectively to both environmental and technological changes (Lee and Pilkington 2017; Rajala and Hautala-Kankaanpää 2023). This illustrates how the RBV, when integrated with the dynamic capabilities perspective, offers a more nuanced understanding of how firms can strategically leverage disruptive technologies to maintain competitive advantage and achieve superior performance in an ever-evolving business environment (Barney 1991; Nowotny et al. 2018).

Lastly, our findings provided empirical support for the mediating role of blockchain technology adoption in the relationship between technological capability and firm performance. Using the dynamic capabilities perspective, technological capability in the context of electronics companies reflects not only their possession of advanced resources, such as technical expertise, infrastructure, and R&D investments, but also their ability to sense, seize, and transform opportunities presented by emerging technologies like blockchain (Teece 2012; Lockett, Thompson, and Morgenstern 2009; Queiroz and Wamba 2019; Rajala and Hautala-Kankaanpää 2023). This dynamic capability enables firms to identify the potential of blockchain in enhancing operational processes, proactively address implementation barriers, and adapt to evolving technological and market demands effectively (Xu et al. 2024). In the context of Bangladesh business context, adopting blockchain technology exemplifies a firm's ability to reconfigure and redeploy its technological resources to maintain competitiveness. From the dynamic capabilities perspective, the interplay between technological capability and blockchain adoption is synergistic and transformative (Teece 2012; S. Zhang et al. 2023). Firms with robust technological capabilities are better equipped to integrate blockchain as a core enabler of their operations, thus unlocking its full potential. Blockchain adoption serves as the mechanism through which technological investments are translated into superior firm performance, such as improved productivity, cost efficiency, and competitive advantage (Nurunnabi 2016; Belhadi et al. 2024). This dynamic interplay emphasizes that the adoption of blockchain technology is not merely additive but serves as a transformative process that reshapes how firms deploy their technological assets to achieve sustained performance in turbulent markets (Teece 2012; Zheng, Luo, and Wang 2014; Bauweraerts and Colot 2023).

Our paper contributes to the current literature. First, many companies are in the early stages of integrating blockchain technologies into their supply chain processes, yet they often struggle with understanding the organizational capabilities required to facilitate successful adoption (Teece 2012; Kim et al. 2011; Al Karim, Rabiul, and Kawser 2023). By employing the RBV framework, our research fills this gap by identifying three key capabilities—innovation capability, information-sharing capability, and technological capability—that are crucial for the effective adoption and utilization of blockchain technology (Barney 1991; Dean and McMullen 2007; Awan et al. 2022). This contribution is particularly valuable as it offers electronics companies a clear, actionable roadmap, grounded in the RBV, outlining how to build and strengthen the internal resources necessary to not only adopt but also maximize the benefits of blockchain (Barney 1991; Zheng, Luo, and Wang 2014; Bahta et al. 2021). It moves beyond merely advocating for the adoption of blockchain by detailing the specific capabilities that must be developed and optimized to ensure its successful implementation (Barney 1991; Bhattacharjee, Jahanshahi, and Chakraborty 2024).

Second, a major challenge many companies face is the hesitation to invest in emerging technologies like blockchain, often opting for traditional methods and established technologies that are perceived as less risky (Uddin, Razzak, and Rahman 2023). Our research makes a significant contribution by addressing this hesitation head-on. By empirically demonstrating the performance implications of blockchain adoption for electronics companies, our study offers robust evidence that can help dispel doubts surrounding the value of this technology (Valdez-Juárez and Castillo-Vergara 2021). This contribution is particularly relevant for firms that are on the fence about whether blockchain is worth the investment. Our findings show that blockchain adoption leads to enhanced firm performance, offering a clear competitive edge, and suggests that reliance on traditional technologies may lead to missed opportunities in innovation, efficiency, and sustainability.

Many companies remain hesitant to adopt blockchain due to concerns over cost, complexity, and uncertain outcomes. However, our research emphasizes blockchain's potential for driving sustainable practices by improving transparency, accountability, and resource efficiency within the supply chain (Torelli and Balluchi 2023). This is particularly important as sustainability becomes a growing imperative for electronics companies globally. Our findings demonstrate that blockchain is not only a performance-enhancing technology but also a critical enabler of environmentally friendly operations. This contribution is valuable for electronics companies grappling with the dual challenge of maintaining profitability while meeting increasing sustainability demands and shows how blockchain can play a pivotal role in balancing these objectives (Wamba, Queiroz, and Trinchera 2020).

6 | Implications of the Study

This study has several important implications for companies that are hesitant to invest in blockchain technology for their supply chains, especially when considering blockchain as an

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environmentally friendly practice. Companies that choose not to invest in blockchain may risk missing out on significant competitive advantages. Our findings demonstrate that blockchain adoption can enhance firm performance by improving transparency, trust, and efficiency within the supply chain. Hesitant firms may be left behind as competitors leverage blockchain to increase operational efficiency, reduce costs, and foster stronger relationships with customers who increasingly demand sustainable practices. This reluctance to innovate may limit their ability to compete in markets where sustainability is becoming a core differentiator.

Traditional supply chain practices, while familiar, may not provide the resilience and adaptability that blockchain enables. Blockchain helps businesses build more resilient and eco-conscious operations by offering greater traceability and accountability. This is especially important in global markets where disruptions (e.g., environmental regulations, supplier misconduct) can have significant operational impacts. Companies that fail to innovate risk having fragile, outdated supply chains that are less capable of responding to modern sustainability challenges. In addition, as blockchain technology becomes more widely adopted in Bangladesh and rest of the world, industry standards may evolve to integrate blockchain-based processes as the norm for transparency and sustainability. Firms that continue using traditional methods could struggle to meet these new standards, particularly if blockchain becomes a requirement in contracts or compliance regulations. Early adopters will have a smoother transition and deeper integration of the technology, leaving latecomers with more significant challenges to overcome.

Furthermore, our study provides important insights for SMEs in Bangladesh, highlighting the strategic importance of integrating organizational capabilities to enhance performance and address environmental concerns. By practicing organizational capabilities, SMEs can meet the evolving expectations of eco-conscious consumers and improve their operational efficiency. The study encourages SMEs to embed organizational capabilities into their strategies for sustainable development. The theoretical framework is grounded in the RBV theory, suggesting that unique resources like advanced supply chain practices and organizational capabilities improve firm performance.

The findings emphasize the need for SMEs to invest in technology and foster a collaborative, digitally empowered environment. Enhancing innovation capability, information-sharing capability, and technological capabilities can help SMEs adapt to changing market demands. The study emphasize that blockchain technology can disruptively mediate the relationships between innovation capability, information-sharing capability, and technological capability, thereby improving firm performance and promoting sustainability. Therefore, SMEs should actively explore blockchain implementation to strengthen their

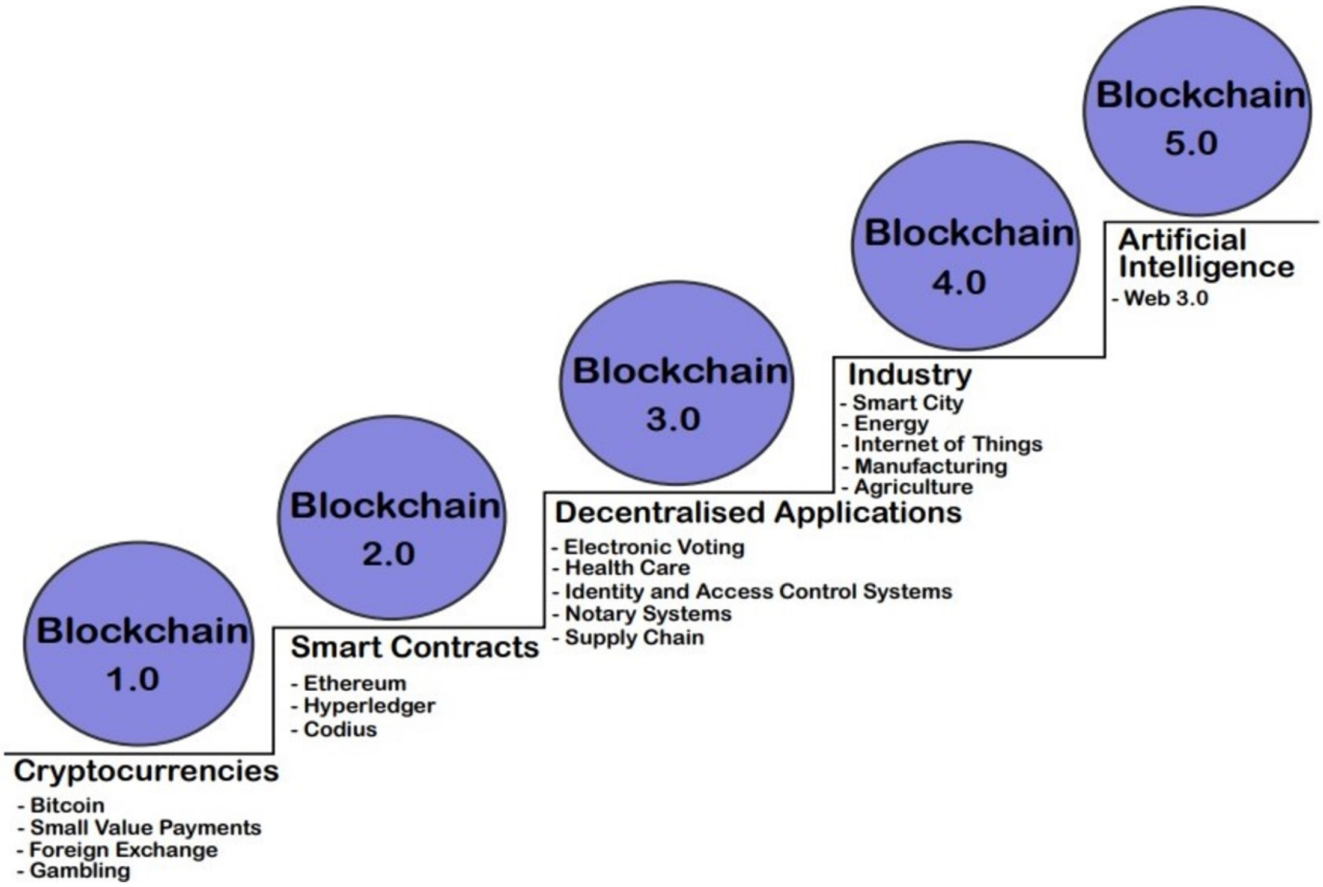

FIGURE 3 | Blockchain evolution (Hameed, Hussain, and Khan 2022).

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supply chain transparency and accountability. A key recommendation is for SMEs to develop an innovation-driven organizational culture, supported by leadership committed to technological advancement. Progressive leadership is critical for encouraging staff to adopt new technologies and sustainable practices. Policymakers and industry support systems should create favorable conditions through laws, financial incentives, and capacity-building programs, tailored to address the specific challenges of SMEs in Bangladesh.

TABLE 8 | Blockchain initiative in various countries.

Country	Initiatives
Australia	The National Blockchain Roadmap: progressing towards a blockchain-empowered future focus areas: • Effective, efficient, and appropriate regulation and standards • Skills and capabilities drive innovation • Strong international investment and collaboration
Bangladesh	National Blockchain Strategy guides Bangladesh into a blockchain-enabled nation to create and maintain a resilient and inclusive information infrastructure and to foster innovation
China	• Blockchain is part of China's Technology Strategy • Blockchain-based Services Network (BSN) provides companies with access to the tools develop blockchain-based applications
United Arab Emirates	The Emirates Blockchain Strategy 2021 aims to capitalize on the blockchain technology to transform 50% of government transactions into the blockchain in platform by 2021. Dubai Blockchain Policy's strategic pillars: • Government efficiency • Industry creation • International leadership
Germany	National Blockchain Strategy and focus on finance, crypto assets securities
India	Blockchain: The India Strategy towards "Enabling Ease of Business, Ease of Governance, and Ease of Living"
Singapore	Singapore Blockchain Innovation Programme (SBIP) with allocation of US$8.9 million

Source: Lai et al. (2023).

Ultimately, the study suggests that and organizational capabilities and adopting blockchain can give Bangladeshi SMEs a competitive edge. Firms that position themselves as technologically advanced and environmentally responsible are more likely to gain consumer loyalty and market preference. The practical implications extend beyond theoretical debates, offering SMEs in Bangladesh actionable strategies to improve performance and contribute to a more resilient, sustainable business environment locally and globally.

7 | Limitations and Future Directions

Though there are some obvious limitations and directions for further research, this study highlights the significant association between organizational capabilities and firm performance in SMEs. The study's initial limitation pertains to the findings' applicability to other sectors or geographical areas, as it was focused exclusively on electronics firms in Dhaka, Bangladesh. Moreover, depending solely on survey data presents the risk of response bias and shallow comprehension. Mixed-method techniques may be used in future studies to get deeper understanding. In addition, a thorough grip may be obtained by investigating the contextual elements affecting the organizational capabilities and its effects on various stakeholder groups beyond manufacturers. To maximize its potential advantages and remove adoption hurdles, it would be beneficial to look at the scalability and real-world implementation issues of blockchain technology in the supply chains of SMEs.

8 | Conclusion

This study explores the relationship between organizational capabilities, the mediating role of blockchain technology adoption, and firm performance in electronics companies in Bangladesh. It highlights the importance of integrating organizational capabilities such as innovation capability, information-sharing capability, and technological capability to balance financial goals with environmental responsibilities (Fernando, Rozuar, and Mergeresa 2021; Deng, Gong, and Wang 2024). The findings show that organizational capabilities positively impact firm performance (Ali 2022; Awan et al. 2022). Blockchain technology plays a key mediating role, connecting innovation capability, information-sharing capability, technological capability, and firm performance (Belhadi et al. 2024). The study emphasizes the need for electronics companies to adopt blockchain and organizational capabilities to foster environmental sustainability and improve performance in a resource-constrained, climate-conscious world (Bui et al. 2021) (Figure 3).

This study examines the role of blockchain technology, particularly its role in enhancing organizational capabilities in electronics companies. Blockchain 4.0 integrates blockchain into business processes, enhancing efficiency while reducing the need for intermediaries. By automating and decentralizing supply chains, Blockchain 4.0 supports environmentally friendly business practices, improving organizational

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capabilities operations (Fernando, Rozuar, and Mergeresa 2021). Additionally, Blockchain 5.0, through AI integration, advances sustainable practices by optimizing firm performance, supporting faster, more secure, and eco-friendly processes (Belhadi et al. 2024). The adoption of Blockchain 5.0, such as, in the *Relictum Pro project*, demonstrates its ability to transform SME operations by enhancing efficiency and sustainability. This study underscores the potential of blockchain technologies to balance economic and environmental goals in supply chains (Table 8).

Bangladesh faces challenges in adopting blockchain technology, including limited local talent and fragmented business knowledge. To address this, the country is promoting integration through regulatory frameworks and talent development. Countries like Australia, China, Germany, and Singapore are advancing blockchain in supply chains, each facing issues such as internet connectivity and skilled labor shortages. While developed countries show faster adoption, Bangladesh and India are slowly catching up. These global examples highlight how blockchain adoption is shaped by unique regional contexts, offering benchmarks for Bangladesh to improve green business practices and economic growth.

Acknowledgments

Authors would like to thank the locally trained research assistants' team who cooperated with us in making the data collection procedures smooth.

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Appendix A

Measurement items

Innovation capability:

1. Our organization actively encourages the generation of innovative ideas to improve sustainability and operational efficiency.
2. We allocate resources specifically for developing and implementing innovative technologies like blockchain.
3. Our organization demonstrates the capability to adapt to new technologies and processes for better performance and sustainability.
4. We foster a culture that values experimentation and supports innovative solutions to address industry challenges.

Information-sharing capability:

1. Our organization ensures that relevant information is easily accessible to employees across departments.
2. We prioritize timely sharing of critical information to support effective decision-making.
3. Our organization promotes transparency by sharing accurate and up-to-date information among team members.
4. We use shared data and insights to enhance collaboration and teamwork across functions.

Technological capability:

1. Our organization adopts advanced technologies to streamline operations and enhance efficiency.
2. We have established robust technological infrastructure to support digital and innovation-driven initiatives.
3. Our workforce is equipped with the necessary technological skills to effectively utilize digital tools and systems.
4. We continuously upgrade our technological resources to remain competitive in a digital economy.

Adoption of blockchain technology:

1. I believe our company should implement blockchain technologies in the NEAR future.
2. We are working out/already have an implementing plan with budget for blockchain technologies.
3. Blockchain technology is a reliable way to maintain privacy of employees like me.
4. Blockchain technology will help stakeholders in browsing information specific to their requirements for taking decision-making.
5. Our organization views blockchain technology as a key driver of sustainability and innovation in achieving strategic objectives.

Firm performance:

1. Our organization has experienced financial growth, such as increased revenue and profitability, due to adopting innovative practices like blockchain technology.
2. The implementation of sustainable and innovative technologies has improved our operational efficiency and reduced costs.
3. Our organization's adoption of advanced technologies has strengthened its competitive position in the market.
4. Sustainability-oriented initiatives in our organization have led to increased customer satisfaction and loyalty.
5. Our firm's performance includes contributing to environmental sustainability and fulfilling corporate social responsibility goals.

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